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Description

This week, I revised the main stages of the data science workflow, including data preprocessing, exploratory data analysis (EDA), feature engineering, supervised learning, and model evaluation. I reviewed how summary statistics and visualisations help identify patterns, relationships, and data quality issues before modelling.

Feelings

Initially, I felt comfortable with the topics because I had studied them before. However, I realised that remembering concepts was easier than applying them independently in Python. This made me recognise the need for more practical experience and greater confidence when solving unfamiliar problems.

Evaluation

The revision helped me connect concepts learned throughout the semester and understand how different stages of the workflow relate to each other. However, covering many topics in a limited time meant I could not practise every technique in depth. I also found unfamiliar problems more challenging than reviewing existing examples.

Analysis

I learned that reliable modelling depends on data quality, appropriate features, and suitable evaluation metrics. For example, accuracy can be misleading when classification classes are imbalanced, while R^2 is commonly used for regression. A model may appear effective under one metric but still perform poorly in practice. This showed me that understanding why a method is appropriate is just as important as knowing how to implement it.

Conclusion

Overall, I have developed a clearer understanding of the data science workflow and the reasons behind different modelling choices. However, I still need more practice applying these concepts independently, particularly when working with unfamiliar datasets.

Action Plan

I will complete two Python exercises each week involving data cleaning, feature engineering, or model evaluation. After each exercise, I will record coding errors, explain my method choices, and identify areas for improvement. I will gradually practise with unfamiliar datasets to assess whether I can apply the techniques independently and strengthen my analytical skills.